Pipe Bend Thickness Report

ASME B31.3-2018 para. 304.2.1 | Pipe bend under internal pressure

Project	QA — مثال كوع
Prepared by	Device preview
Design reference	B31.3-2018 QA
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01 BEND GEOMETRY

Material / grade	User coefficients - example only
Outside diameter D	114.3 mm
Bend radius to centreline R1	171.45 mm (1.5 D)
Inspection / pipe reference	QA-UT

02 DESIGN INPUTS

Internal design gauge pressure P	20 bar(g) = 2 MPa
Allowable stress S	100 MPa
Quality factor E / weld factor W	0.85 / 0.9
Coefficient Y	0.4
Corrosion / erosion allowance	1.5 / 0.5 mm
Mechanical removal allowance	1 mm (before fabrication)

03 CALCULATION TRACE

t = P x D / [2 x ((S x E x W / I) + P x Y)], where I depends on the bend geometry: I_intrados = [4(R1/D) - 1] / [4(R1/D) - 2] and I_extrados = [4(R1/D) + 1] / [4(R1/D) + 2]. I_intrados is always above 1 and governs.

Intrados factor I	1.25
Extrados factor I	0.875
Pressure thickness at intrados	1.8435 mm
Pressure thickness at extrados	1.2955 mm
Matching straight pipe, I = 1	1.4787 mm
Bend wall before fabrication	4.8435 mm
Required remaining wall at intrados	3.8435 mm

04 MEASURED INTRADOS WALL

Minimum measured intrados wall	4 mm
Required remaining wall	3.8435 mm
Measured minus required	0.1565 mm

MEETS the internal-pressure wall requirement for these inputs.

05 LIMITS AND REFERENCES

Basis: ASME B31.3-2018, 304.2.1. Verify user-supplied S, E, W, Y and allowances against the code and project specification. This report checks required bend wall; it does not predict forming or geometric thinning. Excluded: miter bends (304.2.3), external pressure, other sustained/occasional loads, fatigue and remaining-life assessment.